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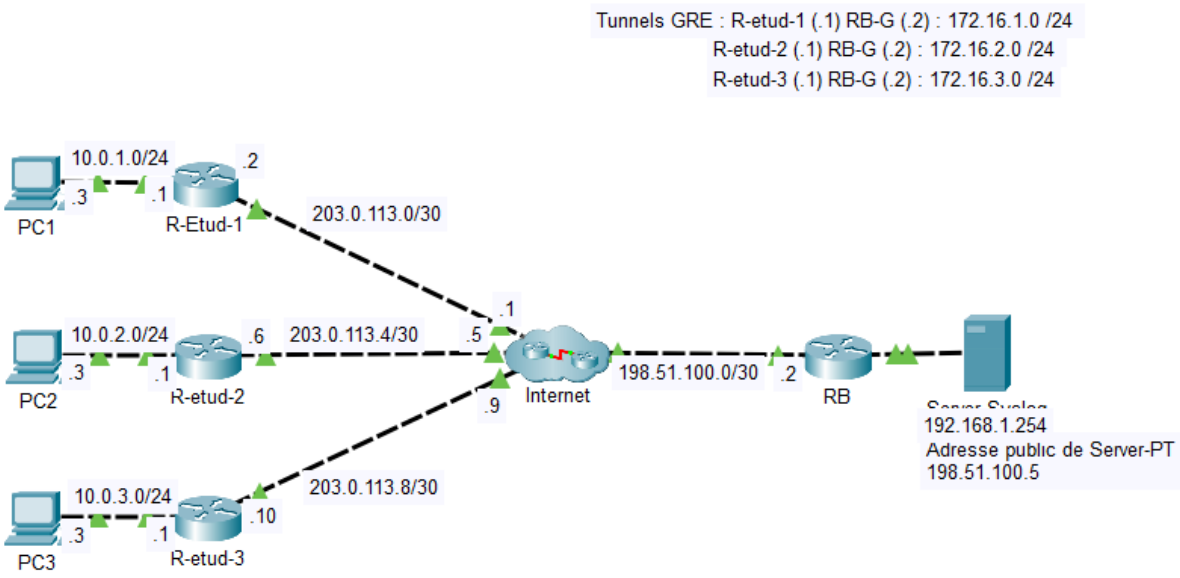
Labo VPN



ip r sur pc1 :

```
root@podd-1:~# ip r
default via 10.0.1.1 dev eth0
10.0.1.0/24 dev eth0 proto kernel scope link src 10.0.1.3
root@podd-1:~# ip a
```

ip a sur pc1 :

```
root@podd-1:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if44: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group
default qlen 1000
    link/ether a6:cc:b1:3f:e9:e6 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.0.1.3/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::a4cc:b1ff:fe3f:e9e6/64 scope link
       valid_lft forever preferred_lft forever
root@podd-1:~#
```

ip r sur pc2 :

```
root@podd-2:~# ip r
default via 10.0.2.1 dev eth0
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.3
root@podd-2:~#
```

ip a sur pc2 :

```
root@podd-2:~# ip r
default via 10.0.2.1 dev eth0
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.3
root@podd-2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if48: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 1a:1a:a2:55:da:90 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.0.2.3/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::181a:a2ff:fe55:da90/64 scope link
       valid_lft forever preferred_lft forever
root@podd-2:~#
```

ip r sur pc3 :

```
root@podd-3:~# ip r
default via 10.0.3.1 dev eth0
10.0.3.0/24 dev eth0 proto kernel scope link src 10.0.3.3
root@podd-3:~#
```

ip a sur pc3 :

```
root@podd-3:~# ip r
default via 10.0.3.1 dev eth0
10.0.3.0/24 dev eth0 proto kernel scope link src 10.0.3.3
root@podd-3:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if80: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 2e:2f:f9:3c:19:e2 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.0.3.3/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::2c2f:f9ff:fe3c:19e2/64 scope link
       valid_lft forever preferred_lft forever
root@podd-3:~#
```

ping depuis PC1 vers la passerelle Locale

Vérifie que le PC1 communique bien avec son routeur local (R-Etud-1).

```
root@podd-1:~# ping 10.0.1.1
PING 10.0.1.1 (10.0.1.1) 56(84) bytes of data.
64 bytes from 10.0.1.1: icmp_seq=1 ttl=255 time=0.385 ms
64 bytes from 10.0.1.1: icmp_seq=2 ttl=255 time=0.378 ms
64 bytes from 10.0.1.1: icmp_seq=3 ttl=255 time=0.312 ms
64 bytes from 10.0.1.1: icmp_seq=4 ttl=255 time=0.379 ms
64 bytes from 10.0.1.1: icmp_seq=5 ttl=255 time=0.343 ms
64 bytes from 10.0.1.1: icmp_seq=6 ttl=255 time=0.349 ms
^C
--- 10.0.1.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5109ms
rtt min/avg/max/mdev = 0.312/0.357/0.385/0.025 ms
root@podd-1:~#
```

ping depuis PC1 vers Serveur Syslog

Test Spoke-to-Hub : Prouve que le tunnel GRE/IPsec vers le siège fonctionne.

```
root@podd-1:~# ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=62 time=40.6 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=62 time=29.3 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=62 time=28.4 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=62 time=27.9 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=62 time=28.0 ms
^C
--- 192.168.1.254 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 27.913/30.855/40.583/4.889 ms
root@podd-1:~#
```

ping depuis PC1 vers PC2

Test Spoke-to-Spoke : Prouve que le routage entre filiales via le Hub est opérationnel.

```
root@podd-1:~# ping 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=61 time=32.1 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=61 time=62.1 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=61 time=52.7 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=61 time=61.6 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=61 time=60.5 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=61 time=59.8 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=61 time=37.9 ms
^C
--- 10.0.2.3 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6008ms
rtt min/avg/max/mdev = 32.090/52.383/62.063/11.477 ms
root@podd-1:~#
```

ping depuis PC1 vers PC3 :

Test Spoke-to-Spoke : Confirme la connectivité totale vers la troisième filiale.

```
root@podd-1:~# ping 10.0.3.3
PING 10.0.3.3 (10.0.3.3) 56(84) bytes of data.
64 bytes from 10.0.3.3: icmp_seq=1 ttl=61 time=45.4 ms
64 bytes from 10.0.3.3: icmp_seq=2 ttl=61 time=33.5 ms
64 bytes from 10.0.3.3: icmp_seq=3 ttl=61 time=42.2 ms
64 bytes from 10.0.3.3: icmp_seq=4 ttl=61 time=41.2 ms
64 bytes from 10.0.3.3: icmp_seq=5 ttl=61 time=40.7 ms
64 bytes from 10.0.3.3: icmp_seq=6 ttl=61 time=39.9 ms
64 bytes from 10.0.3.3: icmp_seq=7 ttl=61 time=38.8 ms
^C
--- 10.0.3.3 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 33.486/40.239/45.353/3.359 ms
root@podd-1:~#
```

ping depuis PC1 vers RB

Vérifie que l'extrémité du tunnel GRE est joignable au niveau IP.

```
root@podd-1:~# ping 172.16.1.2
PING 172.16.1.2 (172.16.1.2) 56(84) bytes of data.
64 bytes from 172.16.1.2: icmp_seq=1 ttl=254 time=21.9 ms
64 bytes from 172.16.1.2: icmp_seq=2 ttl=254 time=31.1 ms
64 bytes from 172.16.1.2: icmp_seq=3 ttl=254 time=19.9 ms
64 bytes from 172.16.1.2: icmp_seq=4 ttl=254 time=17.1 ms
64 bytes from 172.16.1.2: icmp_seq=5 ttl=254 time=16.1 ms
64 bytes from 172.16.1.2: icmp_seq=6 ttl=254 time=14.5 ms
64 bytes from 172.16.1.2: icmp_seq=7 ttl=254 time=12.9 ms
^C
--- 172.16.1.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6008ms
rtt min/avg/max/mdev = 12.883/19.073/31.137/5.684 ms
root@podd-1:~#
```

ping depuis PC2 vers La passerelle Locale :

Valide que PC2 communique avec son routeur local (R-etud-2).

```
root@podd-2:~# ping 10.0.2.1
PING 10.0.2.1 (10.0.2.1) 56(84) bytes of data.
64 bytes from 10.0.2.1: icmp_seq=1 ttl=255 time=0.660 ms
64 bytes from 10.0.2.1: icmp_seq=2 ttl=255 time=0.851 ms
64 bytes from 10.0.2.1: icmp_seq=3 ttl=255 time=0.849 ms
64 bytes from 10.0.2.1: icmp_seq=4 ttl=255 time=0.869 ms
64 bytes from 10.0.2.1: icmp_seq=5 ttl=255 time=0.676 ms
64 bytes from 10.0.2.1: icmp_seq=6 ttl=255 time=0.756 ms
^C
--- 10.0.2.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5120ms
rtt min/avg/max/mdev = 0.660/0.776/0.869/0.085 ms
root@podd-2:~#
```

ping depuis PC2 vers serveur syslog :

Test Spoke-to-Hub : Confirme que le Tunnel 2 vers le siège est opérationnel et sécurisé.

```
root@podd-2:~# ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=62 time=40.2 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=62 time=31.5 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=62 time=29.4 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=62 time=28.8 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=62 time=26.7 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=62 time=24.8 ms
64 bytes from 192.168.1.254: icmp_seq=7 ttl=62 time=23.0 ms
^C
--- 192.168.1.254 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 22.972/29.203/40.232/5.230 ms
root@podd-2:~#
```

ping depuis PC2 vers PC1

Test Spoke-to-Spoke : Prouve que PC2 peut joindre la Filiale 1 via le Hub (RB).

```
root@podd-2:~# ping 10.0.1.3
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.
64 bytes from 10.0.1.3: icmp_seq=1 ttl=61 time=32.5 ms
64 bytes from 10.0.1.3: icmp_seq=2 ttl=61 time=62.5 ms
64 bytes from 10.0.1.3: icmp_seq=3 ttl=61 time=61.8 ms
64 bytes from 10.0.1.3: icmp_seq=4 ttl=61 time=40.9 ms
64 bytes from 10.0.1.3: icmp_seq=5 ttl=61 time=39.3 ms
64 bytes from 10.0.1.3: icmp_seq=6 ttl=61 time=37.6 ms
^C
--- 10.0.1.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 32.533/45.762/62.489/11.875 ms
root@podd-2:~#
```

ping depuis PC2 vers PC3

Test Spoke-to-Spoke : Prouve que PC2 peut joindre la Filiale 3 via le Hub (RB).

```
root@podd-2:~# ping 10.0.3.3
PING 10.0.3.3 (10.0.3.3) 56(84) bytes of data.
64 bytes from 10.0.3.3: icmp_seq=1 ttl=61 time=42.4 ms
64 bytes from 10.0.3.3: icmp_seq=2 ttl=61 time=61.7 ms
64 bytes from 10.0.3.3: icmp_seq=3 ttl=61 time=40.2 ms
64 bytes from 10.0.3.3: icmp_seq=4 ttl=61 time=39.9 ms
64 bytes from 10.0.3.3: icmp_seq=5 ttl=61 time=38.9 ms
64 bytes from 10.0.3.3: icmp_seq=6 ttl=61 time=58.6 ms
^C
--- 10.0.3.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 38.861/46.929/61.664/9.438 ms
root@podd-2:~#
```

ping depuis PC2 vers Interface Tunnel RB :

Vérifie la connectivité IP directe avec l'extrémité du tunnel sur RB.

```
root@podd-2:~# ping 172.16.2.2
PING 172.16.2.2 (172.16.2.2) 56(84) bytes of data.
64 bytes from 172.16.2.2: icmp_seq=1 ttl=254 time=22.0 ms
64 bytes from 172.16.2.2: icmp_seq=2 ttl=254 time=31.2 ms
64 bytes from 172.16.2.2: icmp_seq=3 ttl=254 time=29.7 ms
64 bytes from 172.16.2.2: icmp_seq=4 ttl=254 time=28.2 ms
64 bytes from 172.16.2.2: icmp_seq=5 ttl=254 time=16.7 ms
64 bytes from 172.16.2.2: icmp_seq=6 ttl=254 time=25.2 ms
64 bytes from 172.16.2.2: icmp_seq=7 ttl=254 time=24.2 ms
^C
--- 172.16.2.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 16.651/25.316/31.184/4.615 ms
root@podd-2:~#
```

ping depuis PC3 vers La passerelle Locale :

Vérifie que PC3 communique avec son routeur local (R-etud-3).

```
root@podd-3:~# ping 10.0.3.1
PING 10.0.3.1 (10.0.3.1) 56(84) bytes of data.
64 bytes from 10.0.3.1: icmp_seq=1 ttl=255 time=0.742 ms
64 bytes from 10.0.3.1: icmp_seq=2 ttl=255 time=0.785 ms
64 bytes from 10.0.3.1: icmp_seq=3 ttl=255 time=0.760 ms
64 bytes from 10.0.3.1: icmp_seq=4 ttl=255 time=0.776 ms
64 bytes from 10.0.3.1: icmp_seq=5 ttl=255 time=0.767 ms
64 bytes from 10.0.3.1: icmp_seq=6 ttl=255 time=0.785 ms
^C
--- 10.0.3.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5122ms
rtt min/avg/max/mdev = 0.742/0.769/0.785/0.015 ms
root@podd-3:~#
```

ping depuis PC3 vers Serveur Syslog :

Test Spoke-to-Hub : Confirme que le Tunnel 3 vers le siège est actif et chiffré..

```
root@podd-3:~# ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=62 time=37.2 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=62 time=26.0 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=62 time=24.5 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=62 time=23.3 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=62 time=32.4 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=62 time=31.3 ms
64 bytes from 192.168.1.254: icmp_seq=7 ttl=62 time=29.7 ms
^C
--- 192.168.1.254 ping statistics ---
8 packets transmitted, 7 received, 12.5% packet loss, time 7010ms
rtt min/avg/max/mdev = 23.302/29.209/37.192/4.563 ms
root@podd-3:~#
```

ping depuis PC3 vers PC1 :

Test Spoke-to-Spoke : Prouve que PC3 traverse RB pour atteindre la Filiale 1.

```
root@podd-3:~# ping 10.0.1.3
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.
64 bytes from 10.0.1.3: icmp_seq=1 ttl=61 time=38.3 ms
64 bytes from 10.0.1.3: icmp_seq=2 ttl=61 time=37.6 ms
64 bytes from 10.0.1.3: icmp_seq=3 ttl=61 time=48.2 ms
64 bytes from 10.0.1.3: icmp_seq=4 ttl=61 time=57.2 ms
64 bytes from 10.0.1.3: icmp_seq=5 ttl=61 time=56.7 ms
64 bytes from 10.0.1.3: icmp_seq=6 ttl=61 time=34.7 ms
64 bytes from 10.0.1.3: icmp_seq=7 ttl=61 time=52.2 ms
^C
--- 10.0.1.3 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 34.732/46.438/57.233/8.779 ms
root@podd-3:~#
```

ping depuis PC3 vers PC2

Test Spoke-to-Spoke : Prouve que PC3 traverse RB pour atteindre la Filiale 2.

```
root@podd-3:~# ping 10.0.2.3
PING 10.0.2.3 (10.0.2.3) 56(84) bytes of data.
64 bytes from 10.0.2.3: icmp_seq=1 ttl=61 time=40.2 ms
64 bytes from 10.0.2.3: icmp_seq=2 ttl=61 time=39.1 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=61 time=38.4 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=61 time=37.1 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=61 time=36.8 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=61 time=36.3 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=61 time=35.4 ms
^C
--- 10.0.2.3 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6007ms
rtt min/avg/max/mdev = 35.414/37.608/40.177/1.548 ms
root@podd-3:~#
```

ping depuis PC3 vers RB /Serveur Syslog

Valide la liaison IP point-à-point dans le tunnel GRE n°3.

```
root@podd-3:~# ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=62 time=49.9 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=62 time=39.0 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=62 time=37.8 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=62 time=36.4 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=62 time=35.4 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=62 time=34.1 ms
^C
--- 192.168.1.254 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5007ms
rtt min/avg/max/mdev = 34.068/38.779/49.928/5.231 ms
root@podd-3:~#
```

Show crypto isakmp sa sur Etud-3

Ce test confirme que les routeurs ont réussi à s'authentifier mutuellement avec le mot de passe partagé (`key`) et qu'ils ont créé un canal de contrôle sécurisé. Si l'état est différent, le VPN est "mort".

```
R-Etud-3#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
198.51.100.2    203.0.113.10    QM_IDLE           1001 ACTIVE

IPv6 Crypto ISAKMP SA

R-Etud-3#
```

Show crypto isakmp sa sur RB

C'est la preuve que les données (ton ping) sont transformées en code secret. Si tu pingues et que ces chiffres ne montent pas, les données circulent "en clair" ou sont bloquées par l'ACL.

```
RB-D#show crypto isakmp sa
dst             src             state          conn-id slot status
198.51.100.2    203.0.113.10    QM_IDLE              3    0 ACTIVE
198.51.100.2    203.0.113.2     QM_IDLE              1    0 ACTIVE
198.51.100.2    203.0.113.6     QM_IDLE              2    0 ACTIVE

RB-D#
```

Show ip inter br sur Etud-3

Cela vérifie que le "câble virtuel" entre la filiale et le siège est branché. Si c'est `up / down`, c'est que l'adresse de destination du tunnel est injoignable sur Internet.

```
R-Etud-3#show ip inter br
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            203.0.113.10    YES manual up                    up
FastEthernet0/1            10.0.3.1        YES manual up                    up
Serial0/0/0                unassigned      YES unset  administratively down down
Serial0/0/1                unassigned      YES unset  administratively down down
Tunnel3                    172.16.3.1      YES manual up                    up
R-Etud-3#
```

Show ip ospf neighbor sur RB

Cela prouve que les routeurs se font confiance et s'échangent leurs réseaux privés. Si ce test réussit, le routage automatique est actif.

```
RB-D#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:37    172.16.3.1      Tunnel3
2.2.2.2           0   FULL/  -        00:00:37    172.16.2.1      Tunnel2
1.1.1.1           0   FULL/  -        00:00:36    172.16.1.1      Tunnel1
RB-D#
```

Show ip route sur Etud-3

Cela confirme que le routeur a appris le chemin pour aller chez les autres filiales via le tunnel. Sans le "O", le routeur ne sait pas où envoyer le paquet.

```
R-Etud-3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     203.0.113.0/30 is subnetted, 1 subnets
C       203.0.113.8 is directly connected, FastEthernet0/0
     172.16.0.0/24 is subnetted, 3 subnets
O       172.16.1.0 [110/12111] via 172.16.3.2, 00:48:40, Tunnel3
O       172.16.2.0 [110/12111] via 172.16.3.2, 00:48:40, Tunnel3
C       172.16.3.0 is directly connected, Tunnel3
     10.0.0.0/24 is subnetted, 3 subnets
O       10.0.2.0 [110/12112] via 172.16.3.2, 00:42:40, Tunnel3
C       10.0.3.0 is directly connected, FastEthernet0/1
O       10.0.1.0 [110/12112] via 172.16.3.2, 00:43:30, Tunnel3
O    192.168.1.0/24 [110/1010] via 172.16.3.2, 00:48:40, Tunnel3
S*   0.0.0.0/0 is directly connected, FastEthernet0/0
R-Etud-3#
```

Show crypto ipsec sa sur RB (Tunnel 1)

```
RB-D#sh crypto ipsec sa

interface: Tunnel1
    Crypto map tag: Tunnel1-head-0, local addr 198.51.100.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   current_peer 203.0.113.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 640, #pkts encrypt: 640, #pkts digest: 640
    #pkts decaps: 653, #pkts decrypt: 653, #pkts verify: 653
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 198.51.100.2, remote crypto endpt.: 203.0.113.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0xE5D907B0(3856205744)

     inbound esp sas:
      spi: 0xD064C0B5(3496263861)
        transform: esp-aes esp-sha-hmac ,
```

Show crypto ipsec sa sur RB (Tunnel 2)

```
interface: Tunnel2
    Crypto map tag: Tunnel2-head-0, local addr 198.51.100.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   current_peer 203.0.113.6 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 503, #pkts encrypt: 503, #pkts digest: 503
    #pkts decaps: 519, #pkts decrypt: 519, #pkts verify: 519
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

     local crypto endpt.: 198.51.100.2, remote crypto endpt.: 203.0.113.6
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0xF279DA88(4068072072)

     inbound esp sas:
      spi: 0x3F79D3C0(1064948672)
 --More--
```

Show crypto ipsec sa sur RB (Tunnel 3)

```
interface: Tunnel3
    Crypto map tag: Tunnel3-head-0, local addr 198.51.100.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   current_peer 203.0.113.10 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 438, #pkts encrypt: 438, #pkts digest: 438
    #pkts decaps: 454, #pkts decrypt: 454, #pkts verify: 454
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 198.51.100.2, remote crypto endpt.: 203.0.113.10
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0xCA47CBFA(3393702906)

     inbound esp sas:
 --More--
```